

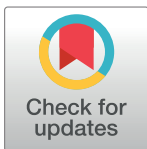
RESEARCH ARTICLE

Examination of water quality at the household level and its association with diarrhea among young children in Ghana: Results from UNICEF-MICS6 survey

Mohammed Husein¹, Carole Debora Nounkeu², Seth Armah^{3*}, Jigna M. Dharod³

1 Department of Nutrition and Food Science, University of Ghana, Accra, Ghana, **2** Limbe Regional Hospital, Limbe, Cameroon, **3** Department of Nutrition, University of North Carolina at Greensboro, Greensboro, NC, United States of America

* s_armah@uncg.edu



OPEN ACCESS

Citation: Husein M, Nounkeu CD, Armah S, Dharod JM (2023) Examination of water quality at the household level and its association with diarrhea among young children in Ghana: Results from UNICEF-MICS6 survey. PLOS Water 2(6): e0000049. <https://doi.org/10.1371/journal.pwat.0000049>

Editor: Vikram Kapoor, University of Texas at San Antonio, UNITED STATES

Received: June 14, 2022

Accepted: May 22, 2023

Published: June 13, 2023

Copyright: © 2023 Husein et al. This is an open access article distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

Data Availability Statement: We used publicly available Ghana UNICEF-MICS6 survey. UNICEF MICS. (2022). MICS6 TOOLS. Retrieved 6 1, 2022, from UNICEF: <https://mics.unicef.org/tools#datacollection>.

Funding: The authors received no specific funding for this work.

Competing interests: The authors have declared that no competing interests exist.

Abstract

Ghana has made significant progress in expanding water services, but microbial contamination of water is still a major public health problem. The objectives of the study were to: 1) Examine sociodemographic and water access related predictors for the point of use or drinking water quality among rural and urban households, and; 2) Determine the association between the point of use water quality and prevalence of diarrhea among young children in rural and urban households. A secondary data analysis was carried out using the Ghana UNICEF-MICS6 survey taking into account the complex survey design. Logistic regression models were used to carry out the objectives. Among the 2317 households included for water quality testing, majority reported using improved source of drinking water. However, use of unimproved source of water was more common among rural households. In examining water quality at the point of use, it was found that more than 60% of the samples had mid to high levels of *E. coli* count, with significantly more common among rural compared to urban households ($p < .0001$). The prevalence of diarrhea among under 5 children was 16.9%. In estimating the risk, *E. coli* count was not associated with higher diarrhea prevalence. However in urban areas, water storage was associated with increased risk of caregiver-reported diarrhea in children. Also, other factors such as child's age, maternal education, region and household wealth index predicted diarrhea prevalence. In Ghana, contaminated point of use drinking water is more common in rural household, and in urban areas, water storage is associated with increased risk of caregiver-reported diarrhea in children. In the future, investigation of living condition and environmental hygiene is warranted to further understand different pathways through which risk of diarrhea increases among children.

Introduction

Safe and clean water is critical to sustainable development and for human survival. It is recognized that access to clean and safe water is a right of every human being [1]. However, according to the progress report on the Sustainable Development Goal (SDG) # 6 on access to safe and clean water for all, about 2 billion people or 26% of the world's population lack access to safely managed drinking water services [2]. It is recognized that lack of universal access to clean water is impeding the progress in meeting nutrition and public health goals in low- and middle-income countries.

Based on the WHO/UNICEF's Joint Monitoring Program indicators, a drinking water source is considered improved if it is protected from contamination by nature of its construction or through active intervention [2]. Hence, improved water source types include piped water, standpipe, borehole, and protected dug well. However, a review of the water quality studies highlights that getting water from an improved source does not guarantee clean and safe point of use water. Several factors, including water storage facility at the household level, availability of improved sanitation at the community level, and condition of the improved water source affect the safety and quality of water [3]. The 2022 WHO/UNICEF's JMP report indicated that access to an improved water source is more common in urban compared to rural areas in Sub-Saharan Africa [2]. Especially studies in the rural areas show that not only the type of water source (improved vs. unimproved) but the distance to the water source and related water fetching burden, such as, collecting and storing water, type of water container and proximity to livestock affect the point of use water quality [4–6].

Understanding the sources and pathways through which fecal contamination occurs in the drinking or point of use water is critical in planning concerted efforts and meeting SDG # 6. The achievement of this goal will allow meeting other SDG targets too, such as ending preventable deaths of newborns and children under 5 years of age, since unsafe water is the major risk factor for diarrheal diseases among children causing high levels of morbidity and mortality [7]. In the study involving the UNICEF-MICS data from 27 countries, it was found that low income and rural households were at significant increased risk of using contaminated drinking water than their counterparts, however, the relationship between water quality and diarrhea risk among children was not explored [8].

In Ghana, access to water has improved significantly in the past decade, but microbial contamination of drinking water is still an issue in the country. According to the latest Ghana-MICS report, *Escherichia coli* (*E. coli*) contamination in drinking water is more common than in the source water [9]. Hence, to understand water quality issues at the household level and whether disparities exist between rural and urban households, the objectives of the study were to utilize the latest national level data to: 1) Examine sociodemographic and water access related predictors for the point of use or drinking water quality among rural and urban households, and; 2) Determine the association between the point of use water quality and prevalence of diarrhea among young children in rural and urban households.

Methods

To meet the study objectives, UNICEF-MICS 2018 dataset for Ghana was used. Ghana is a country in the western part of Africa with a population of about 30 million. The country is currently divided into 16 regions. However, at the time of the collection of the MICS 6 data, there were only 10 regions in Ghana. Thus, our analysis presents the results for the original 10 regions. The Ghana UNICEF-MICS 2018 was designed to be useful in assessing multiple indicators relating to women and children at various levels including national, regional and area (rural/urban) levels. The fieldwork was done from October 15th, 2017 to January 15th, 2018,

representing mostly the dry season in Ghana. In the design, urban and rural areas designated by the government within each of the 10 regions were used as the main sampling strata, and households were selected using a two-stage sampling design with a probability approach. Oversampling was done for households with women within the age range of 20–24 years to improve the precision for estimates of indicators of early marriage. Due to the nature of the sampling, sample weights were provided in the data to ensure accurate estimates when reporting results. A total of 3219 rural and urban households were involved in water quality testing. One of the children under 5 from the water quality testing households was selected to assess the diarrhea prevalence and its association with the water quality at the household level. To test the objectives, we selected water quality testing households that had at least one under 5 child which resulted in a sample 2317 households. We used the household data (hh.sav) and the children under 5 data (ch.sav) files. From the household data file, we used the following variables: household wealth index, household head's education, mother's or caretaker's education, urban vs rural household, and region. The computed wealth index and its categories representing a proxy measure of the long-term standard of living of the household was utilized for household income status. The water access related variables and drinking water quality test results were also extracted from household data file. The water access related variables used were: improved vs unimproved drinking water source, location (in the vicinity or not), if not in the vicinity then distance and time to water source, water treatment (yes or no) and water storage (yes or no). In case of water quality measurement, details can be found in the survey manual [10]. But, a range of quality control measures were put in place to ensure accuracy in measurement. For instance, "blank" samples known to be free of contamination were tested to ensure that contamination in the water samples was not due to poor survey testing techniques, such as dirty hands or equipment. Training was also carried out to ensure accuracy and robustness of the water quality data. *E. coli* count was reported in the range of 0 to 100 cfu. The water testing results of > 100 cfu per 100 mL sample water was grouped as one category of >100 cfu. For this reason, we grouped households into four categories (0 cfu or absent, 1–10 cfu or low, 11–100 cfu or medium and >100 cfu). From the children under 5 data file, child's sex, age and reported prevalence of diarrhea in the past 2 weeks were retrieved.

Statistical analysis

Data were analyzed using R software. We used the package survey in R [11]) taking into account the strata, principal sampling unit (PSU), and sampling weight. For continuous variables, such as time to water source, we reported the means and the standard error (SE). For categorical variables, the adjusted number (frequencies) and percentages were reported. For each analysis households with missing data were excluded. We used chi-squared analysis with the adjusted Wald statistics to determine the bivariate association between categorical variables, such as area (rural vs. urban) and sociodemographic characteristics of household. To determine sociodemographic and water access related predictors of water quality overall and specific by urban and rural households, we used ordinal logistic regression analysis utilizing 4 *E. coli* count categories as dependent variable. The independent variables were water source, time to water source, water treatment, water storage, caretaker education level, wealth index category of the household, and the region. For prevalence of diarrhea, a binary variable (yes or no) in reference to question on whether the reference child in the household experienced diarrhea in the past two weeks was used. Separate models were constructed for urban and rural areas, and an overall model including both areas was also constructed. In all the regression models, initially we also included the following sanitation variables: 1) Type of toilet facility used (improved vs unimproved vs open defecation) and 2) whether the toilet facility was

shared with other households. Using the WHO/UNICEF JMP ladder for sanitation, the combination of these two variables were used to group households into the following 4 sanitation categories: improved (improved facility and not shared with other households) vs. limited (improved facility shared with other households) vs. unimproved facility vs. open defecation. However, this sanitation ladder did not significantly affect water quality or caregiver-reported diarrhea prevalence in children. Therefore, it was excluded to maintain statistical power. Statistical significance was set at $p \leq 0.05$.

Results

A total of 2317 households from the 2018 Ghana MICS survey were used for this analysis, with approximately 44% of the households being urban households. Approximately 37% of all household heads were educated up to the junior secondary or middle school level. Higher percentage of household heads in the urban compared to rural areas had senior secondary or higher levels of education ($p < 0.0001$; Table 1). Overall, 85% of households used water from improved sources, with 97% and 75% for urban and rural households, respectively. Among children, 48% were males and the mean age was 2.0 ± 0.0 years. As shown in Table 2, approximately 89% of households had their water sources outside their own dwelling or yard. For these households, the average time to get to water source, fetch water and return was 17.0 ± 1.3 minutes. The distance to water was significantly higher in rural households (20.4 ± 1.9 mins) compared to the urban group (10.8 ± 0.7). In terms of *E. coli* contamination, it was absent in 23% of water samples. In comparison, 11% of rural households versus 38% of urban households had no *E. coli* present in drinking water ($p < 0.0001$). Subsequently, as shown in the table medium to high levels of *E. coli* cfu was also more common in rural versus urban households (Table 2).

The determinants of *E. coli* count in drinking water are shown in Table 3. For all households put together, area, water source, water storage and wealth index determined the *E. coli* count. Medium to high levels of *E. coli* was more likely to be common in households that used unimproved water sources ($p < 0.0001$), and those that stored water ($P < 0.0001$). It was also likely to decrease in the fourth and richest quintiles of wealth index compared to the poorer (lowest) quintile ($p < 0.05$). Between, rural and urban households, rural households were more likely to have contaminated drinking water with *E. coli*. Specifically, in rural households, unimproved water sources, and water storage were associated with increased *E. coli* count, while the two highest quintiles of wealth index were associated with decreasing *E. coli* count. In the urban households, only water storage and region were associated with *E. coli* count (Table 3).

In examining diarrhea occurrence in children in the past two weeks, different levels of *E. coli* count were not a significant predictor after controlling for household wealth index and other sociodemographic variables. Overall, prevalence of diarrhea decreased with increase in child's age ($p = 0.001$, Table 4). Specifically, among rural households, child's age, region and wealth index were associated with diarrhea prevalence ($p \leq 0.05$). Whereas in urban areas, children living in western region, mother having a higher education or above were less likely to report child diarrhea incidence in the past two weeks. Among urban households, storing water in the households was also associated with higher likelihood of diarrhea among young children. Households that did not store water were less likely to report diarrhea prevalence in children.

Discussion

The results of this nationally representative study in Ghana indicates that though use of improved drinking water source is common, the point of use water contaminated with *E. coli*

Table 1. Characteristics of Household (HH) by area (Urban Vs Rural)¹.

	Urban (n = 1024)		Rural (n = 1293)		Total (n = 2317)		P
	Number of HH	PERCENTAGE	Number of HH	PERCENTAGE	Number of HH	PERCENTAGE	
HH Wealth Index							
Poorest	58	5.7	450	34.8	508	21.9	<0.0001
Second	119	11.6	350	27.1	469	20.2	
Middle	171	16.7	227	17.6	398	17.2	
Fourth	293	28.6	122	9.4	415	17.9	
Richest	384	37.5	143	11.1	527	22.7	
HH Head Education							
	174	17.0	428	33.1	602	26.0	
Primary	130	12.7	262	20.3	392	16.9	<0.0001
JSS/JHS/Middle	428	41.8	434	33.6	862	37.2	
SSS/SHS/ Secondary	152	14.8	109	8.4	261	11.3	
Higher	140	13.7	59	4.6	199	8.6	
Mother/Caretaker education ³							
Primary or less	169	19.9	314	36.6	483	28.3	
JSS/JHS/Middle	419	49.4	451	52.6	870	51.0	<0.0001
SSS/SHS/ Secondary	159	18.8	67	7.8	226	13.3	
Higher	101	11.9	25	2.9	126	7.4	
Region							
Western	100	9.8	171	13.2	271	11.7	
Central	109	10.6	132	10.2	241	10.4	
Greater Accra	174	17.0	52	4.0	226	9.8	
Volta	35	3.4	138	10.7	173	7.5	
Eastern	94	9.2	132	10.2	226	9.8	<0.0001
Ashanti	330	32.2	245	19.0	575	24.8	
Brong Ahafo	99	9.7	131	10.1	230	9.9	
Northern	59	5.8	194	15.0	253	10.9	
Upper east	17	1.7	56	4.3	73	3.2	
Upper west	8	0.8	41	3.2	49	2.1	

¹All analysis were conducted taking into account the sampling weights, strata and principal sampling units. Sample sizes for different variables may slightly deviate from the total number of 2317 due to rounding or missing data. JSS: Junior Secondary School; JHS: Junior High School; SSS: Senior Secondary School; SHS: Senior High School.

<https://doi.org/10.1371/journal.pwat.0000049.t001>

exist. Prevalence of drinking water or the point of use water contaminated with *E. coli* was 62.3% in urban and 88.7% in rural households. This study also shows that disparities exist in the odds of drinking clean water by region and income level, indicating regional and area level specific investigation and initiatives are needed to improve water quality and bring equity in water security.

Since the beginning of the “Water and sanitation for all” initiative launched by the UN agencies and charitable organizations in the early 1980s, water supply and sanitation infrastructures coverage in middle- and low-income countries has improved significantly [12,13]. However, consumption of contaminated drinking water still represents a significant issue in low and middle countries and represents a huge burden on country’s productivity and economic potential [14,15]. Our study showed that fecal contamination of water was significantly influenced by rural location and poverty. Similarly, a systematic review of 345 studies involving coliform measurement of 133,460 water samples in Africa and South-East Asia, indicated

Table 2. Household water sources and related variables by area (Urban vs Rural)¹.

	Urban (n = 1024)		Rural (n = 1293)		Total (n = 2317)		P
	Number of HH	PERCENTAGE	Number of HH	PERCENTAGE	Number of HH	PERCENTAGE	
Main source of drinking water							
Improved WS	992	96.9	973	75.2	1965	84.8	<0.0001
Unimproved WS	32	3.1	320	24.7	352	15.2	
Where is that water source located							
In own dwelling	9	1.1	6	0.5	15	0.8	0.09
In own yard / plot	120	16.0	80	6.5	200	10.1	
Elsewhere	620	82.8	1139	93.0	1759	89.1	
Is anything done to the water to make it safer to drink?							
Yes	81	7.9	105	8.1	186.00	8.0	0.9
No	943	92.1	1187	91.8	2130	92.0	
Time to water source and back ² (mins)	10.8	0.7	20.4	1.9	17.0	1.3	<0.0001
<i>E. coli</i> count							
0	386	37.7	146	11.3	532	23.0	<0.0001
1–10	180	17.6	148	11.4	328	14.2	
11–100	280	27.4	367	28.4	647	27.9	
>100	177	17.3	632	48.9	809	34.9	
Water stored							
Yes	767	74.9	1183	91.5	1950	84.2	<0.0001
No	257	25.1	110	8.5	367	15.8	
Child had diarrhea in past two weeks							
Yes	178	17.4	213	16.5	391	16.9	0.69
No	846	82.6	1078	83.5	1924	83.1	

¹All analyses were conducted taking into account the sampling weights, strata and principal sampling units.

²Values are means and standard errors instead of number of households and percentage. T-test was used to compare means between urban and rural households. It refers to the households who did not have water source in their own yards or dwellings.

<https://doi.org/10.1371/journal.pwat.0000049.t002>

the rural areas are most affected by fecal contamination [3]. In understanding the water contamination risk, reviews have shown that post-collection contamination is more common and plays a major role in predicting water quality compared to just knowing whether the water source is improved or unimproved [16,17]. Consequently, the estimation of access to safe and clean water based on use of improved water sources has been criticized since it causes an underestimation of the burden of waterborne diseases. For instance, in a meta-analysis of 45 studies to assess the extent of fecal contamination at the source and in household stored water, it was found that whereas 46% of the point of collection samples were contaminated, the percentage of contaminated household water samples was almost double (75%) [16]. To address poor water quality among rural households where water collection from piped systems is uncommon, the use of water treatment techniques, is recognized as a necessary step in ensuring clean water. In this study, the use of treatment methods to purify water was not common. In the analyses of the UNICEF-MICS 2014-Nepal survey, Kandel et al. found that households who boiled water had about 50% lower odds of fecal contamination than households who did not boil water [18]. Further, the results of this study demonstrated that household wealth status was a significant predictor of poor water quality. Households in richest wealth index category were less likely to consume unclean water, this pattern was seen in both rural and urban households.

In comparison by regions in Ghana, in urban areas Volta region had significantly increased risk of contamination compared to Greater Accra (the reference region). Guzmán and Stoler

Table 3. Factors that determine *Escherichia coli* in drinking water at point of use (n = 2317)¹.

	Urban (n = 1024)			Rural (n = 1293)			Total (n = 2317)		
	OR	95% CI	P	OR	95% CI	P	OR	95% CI	P
Intercept (<i>E. coli</i> count)									
Intercept: <i>E. coli</i> count = 0	4.64	(1.34,16.07)	0.02	4.78	(1.04,21.97)	0.04	3.86	(1.66,8.96)	0.002
Intercept: <i>E. coli</i> count = 1–10	1.49	(0.43,5.16)	0.53	1.77	(0.4,7.88)	0.46	1.35	(0.58,3.13)	0.49
Intercept: <i>E. coli</i> count = 11–100	0.4	(0.11,1.41)	0.15	0.35	(0.08,1.53)	0.16	0.31	(0.13,0.72)	0.01
Area (Ref = "Urban")									
HH6RURAL							1.47	(1.02,2.12)	0.04
Water source (Ref = "Improved")									
Unimproved	2.03	(0.81,5.12)	0.13	6.92	(3.61,13.26)	<0.0001	4.95	(2.97,8.25)	<0.0001
Water stored (Ref = "Yes")									
NO	0.23	(0.13,0.42)	<0.0001	0.19	(0.11,0.35)	<0.0001	0.22	(0.14,0.35)	<0.0001
Water Treated (Ref = ("Yes"))									
No	0.95	(0.48,1.87)	0.88	1.68	(0.88,3.22)	0.12	1.27	(0.79,2.05)	0.33
Time to water source and back (mins)	1.01	(0.99,1.02)	0.59	1	(0.99,1.01)	0.75	1	(0.99,1.01)	0.97
Mother/Caretaker Education level (Reference = primary or less)									
JSS/JHS/Middle	1.1	(0.65,1.84)	0.73	0.89	(0.6,1.34)	0.59	0.97	(0.71,1.32)	0.84
SSS/SHS/ Secondary	1.69	(0.79,3.6)	0.17	1.55	(0.83,2.93)	0.17	1.56	(0.92,2.65)	0.10
Higher	1.62	(0.61,4.3)	0.33	0.55	(0.23,1.29)	0.17	1.03	(0.54,1.98)	0.92
Wealth Index (Ref = "Poorer")									
Second	1.47	(0.6,3.57)	0.40	1.02	(0.62,1.67)	0.95	1.08	(0.7,1.66)	0.73
Middle	0.61	(0.26,1.45)	0.26	0.89	(0.53,1.49)	0.66	0.73	(0.48,1.11)	0.15
Fourth	0.46	(0.19,1.12)	0.09	0.49	(0.26,0.92)	0.03	0.47	(0.29,0.76)	0.002
Richest	0.27	(0.11,0.69)	0.01	0.34	(0.13,0.91)	0.03	0.3	(0.16,0.55)	0.0001
Region (Ref = Greater Accra)									
Western	1.73	(0.75,3.98)	0.20	0.54	(0.13,2.2)	0.39	0.78	(0.41,1.48)	0.44
Central	1	(0.44,2.29)	0.99	1.61	(0.4,6.44)	0.50	1.35	(0.72,2.51)	0.35
Volta	2.29	(1.1,4.8)	0.03	1.32	(0.32,5.39)	0.70	1.53	(0.79,2.98)	0.21
Eastern	0.77	(0.38,1.57)	0.48	0.91	(0.23,3.66)	0.89	0.87	(0.49,1.57)	0.65
Ashanti	0.81	(0.4,1.65)	0.56	1.13	(0.25,5.2)	0.87	0.99	(0.52,1.88)	0.97
Brong Ahafo	0.98	(0.43,2.23)	0.96	0.9	(0.22,3.76)	0.89	0.96	(0.51,1.81)	0.90
Northern	1.85	(0.76,4.52)	0.18	1.65	(0.38,7.04)	0.50	1.81	(0.91,3.62)	0.09
Upper East	0.93	(0.25,3.41)	0.91	1.08	(0.26,4.49)	0.92	1.04	(0.51,2.11)	0.92
Upper West	0.81	(0.38,1.74)	0.59	1.35	(0.33,5.44)	0.68	1.14	(0.62,2.08)	0.68

¹All analysis were conducted taking into account the sampling weights, strata and principal sampling units. Sample sizes have been adjusted for sampling weights.

<https://doi.org/10.1371/journal.pwat.0000049.t003>

conducted a study in Ghana to compare water quality in rural and semi-urban communities in several regions including the Volta region [19]. Their results also showed similar regional differences in water quality and concluded that intermittent water shortage and non-functional improved sources caused mixed use of improved and open surface water in certain areas increasing the risk of drinking contaminated water. In recent years, use of the bagged sachet water for drinking has become very common in Ghana in both rural and urban areas [19,20]. In our study, we were not able to specifically identify the use of sachets or bottled water for drinking. But, according to the 2014 Ghana Demographic and Health Survey results, sachet water was the primary drinking water source for 29% of households overall, with about 43% of urban and 12% of rural households reporting its use [21].

Table 4. Predictors of caregiver-reported Diarrheal Morbidity among 5 year old or under Ghanaian Children¹.

	URBAN (n = 1024)				RURAL (n = 1293)			ALL (n = 2317)	
	OR	95% CI	P	OR	95% CI	P	OR	95% CI	P
(Intercept)	0.51	(0.06,4.47)	0.54	0.41	(0.12,1.42)	0.16	0.46	(0.13,1.63)	0.23
<i>E. coli</i> count (Ref = ">100")									
Absent (0)	0.79	(0.27,2.27)	0.65	0.83	(0.34,1.98)	0.67	0.81	(0.42,1.55)	0.52
Low (1–10)	0.81	(0.25,2.67)	0.73	0.62	(0.27,1.47)	0.28	0.8	(0.4,1.61)	0.53
Medium (11–100)	1.03	(0.38,2.78)	0.96	1.04	(0.51,2.15)	0.91	1.12	(0.67,1.89)	0.66
Water source (Ref = "Improved")									
Unimproved	0.25	(0.03,2.44)	0.23	0.73	(0.34,1.57)	0.42	0.74	(0.37,1.47)	0.39
Time to water source and back (mins)	1.01	(0.99,1.02)	0.22	1.01	(1,1.02)	0.22	1.01	(1,1.02)	0.08
Water Treated (Ref = ("Yes"))									
No	1.47	(0.4,5.43)	0.56	1.84	(0.74,4.61)	0.19	1.59	(0.71,3.53)	0.26
Water Stored (Ref = ("Yes"))									
No	0.35	(0.13,0.93)	0.04	1.04	(0.36,3.03)	0.94	0.57	(0.26,1.22)	0.15
Child sex (Ref = "Male")									
Female	1.15	(0.6,2.2)	0.68	1.03	(0.62,1.7)	0.92	1.13	(0.75,1.69)	0.56
Child Age	0.8	(0.63,1)	0.05	0.74	(0.61,0.91)	0.004	0.78	(0.67,0.9)	0.001
Mothers' education (Ref = " ≤ Primary")									
JSS/JHS	0.61	(0.33,1.13)	0.11	1.18	(0.61,2.31)	0.62	0.91	(0.58,1.42)	0.67
SSS/SHS/TECH/VOC/COMM	0.5	(0.16,1.51)	0.21	0.5	(0.16,1.58)	0.23	0.56	(0.24,1.31)	0.18
Higher	0.1	(0.01,0.61)	0.01	1.34	(0.2,9.09)	0.77	0.38	(0.09,1.62)	0.19
Area (Ref = "Urban")									
HH6RURAL							0.67	(0.43,1.05)	0.08
Region (Ref = Greater Accra)									
Western	0.25	(0.06,0.95)	0.04	0.46	(0.16,1.33)	0.15	0.51	(0.23,1.15)	0.11
Central	0.63	(0.19,2.09)	0.44	0.52	(0.18,1.49)	0.22	0.74	(0.33,1.66)	0.47
Volta	0.49	(0.1,2.35)	0.37	0.72	(0.21,2.51)	0.61	0.85	(0.3,2.36)	0.75
Eastern	1.14	(0.34,3.76)	0.83	0.39	(0.11,1.3)	0.12	0.77	(0.32,1.81)	0.54
Ashanti	1.18	(0.41,3.43)	0.75	0.7	(0.25,1.96)	0.49	1.14	(0.53,2.47)	0.73
Brong Ahafo	0.47	(0.09,2.36)	0.36	0.27	(0.08,0.94)	0.04	0.47	(0.17,1.32)	0.15
Northern	1.14	(0.32,4.13)	0.84	1.89	(0.59,6.01)	0.28	1.94	(0.8,4.7)	0.14
Upper East	0.5	(0.08,3.13)	0.45	0.7	(0.23,2.18)	0.54	0.86	(0.33,2.24)	0.75
Upper West	5.52	(0.83,36.82)	0.08	0.39	(0.12,1.26)	0.12	0.65	(0.25,1.69)	0.38
Wealth Index (Ref = "Poorer")									
Second	1.1	(0.25,4.77)	0.90	0.43	(0.21,0.88)	0.02	0.56	(0.3,1.04)	0.07
Middle	1.27	(0.29,5.55)	0.75	0.6	(0.28,1.29)	0.19	0.75	(0.41,1.4)	0.37
Fourth	0.75	(0.17,3.25)	0.70	0.87	(0.35,2.17)	0.76	0.66	(0.32,1.34)	0.25
Richest	0.66	(0.14,3.24)	0.61	0.25	(0.06,0.95)	0.04	0.48	(0.19,1.23)	0.12

¹ Dependent variable was dichotomous (yes or no) based on caregiver-reported prevalence of diarrhea in the past 2 weeks. All analysis were conducted taking into account the sampling weights, strata and principal sampling units. Sample sizes may slightly deviate from the total number of 2317 due to rounding or missing data. OR, odds ratio.

<https://doi.org/10.1371/journal.pwat.0000049.t004>

In our study, no significant association was found between *E. coli* in drinking water and the caregiver-reported prevalence of diarrhea among young children. Similarly, in a meta-analysis of 28 observational and intervention studies, no significant positive relationship has been found between point-of-use water contamination (*E. coli* and thermotolerant coliforms) and diarrhea among children [22]. In the study utilizing Bangladesh UNICEF-MICS, Khan et al

found a significant dose-response association i.e., as the *E. coli* count increased in drinking water the risk of diarrheal episodes among under-5 children also increased [23]. In a prospective longitudinal study, it was found that contaminated drinking or point of use water was associated with subsequent incidence of diarrhea among children [24]. As argued by Levy, measurement of water quality just once with concurrent measurement of diarrhea might mask the link between the two as most diarrhea causing pathogens have incubations periods of > 24 hours [25]. Further, results of the meta-analysis highlights that not only contaminated drinking water but contaminated surfaces such as child's hands play a role in determining risk for diarrhea [26]. While we did not find any association between *E. coli* count and diarrhea prevalence, water storage was associated with increased risk of reported diarrhea in children, particularly in urban areas.

It was also not surprising that maternal education was a significant predictor for caregiver-reported diarrhea among children in our study. Maternal education has consistently remained significant, with children of less-educated mothers shown to be at higher risk of experiencing diarrhea [27]. In a study involving 915 censuses and nationally representative surveys, it was found that between 1970 and 2009, an increase in the average years of education among women contributed 50% reduction in under-5 mortality [28]. Children of more educated mothers are more likely to receive vaccines, visit a doctor for sickness, sleep under insecticide-treated nets, and benefit from other health-protective practices. In our study too, the protective effect of maternal education can also be due to high likelihood of practicing behaviors that help prevent water contamination, such as more likely to treat water, use covered containers to store water, and/or wash hands with soap and water.

Notwithstanding the evidence, there are some limitations to consider while interpreting our findings, as well as for future research. First, the results are based on a momentary snapshot of household water quality. However, evidence suggests that water quality varies significantly depending on the season (dry vs. wet), from day to day based on water services functioning, and can even be conditioned by the time availability of the main water fetcher. Further, secondary data analyses carry inherent limitations, including utilizing only the variables that are reported and there is also possibility of unmeasured and residual confounding affecting the ability to measure the strength of the association between behaviors and outcome variables. This study targeted one major source of contamination i.e. drinking water, but other sources, such as food, children's hands and surroundings, can also cause fecal-oral route transmission. Finally, the use of the ordinal logistic regression model in the analysis of this complex survey data assumes that across all levels of the outcome variable, as we move from one level of to the next, the effect of an independent variable is constant (proportional odds assumption).

Conclusions

Our study results indicate that, disparity in water quality exist in Ghana, with poor water quality more common in rural versus urban households. While we did not find any association between *E. coli* count and diarrhea prevalence, water storage was associated with increased risk of reported diarrhea in children in urban areas. In light of our findings, future investigation on how other factors such as environmental hygiene, living condition, and use of food preparation practices affect diarrhea risk, is warranted. A full understanding of factors and specific pathways that affect water quality at the point of use is also critical in meeting the SDG goals and ensuring reduction in inequity due to lack of access to safe and clean water.

Author Contributions

Conceptualization: Mohammed Husein, Carole Debora Nounkeu, Seth Armah, Jigna M. Dharod.

Data curation: Seth Armah.

Formal analysis: Seth Armah.

Methodology: Mohammed Husein, Carole Debora Nounkeu, Seth Armah, Jigna M. Dharod.

Supervision: Jigna M. Dharod.

Validation: Mohammed Husein, Seth Armah, Jigna M. Dharod.

Visualization: Jigna M. Dharod.

Writing – original draft: Mohammed Husein, Carole Debora Nounkeu, Seth Armah, Jigna M. Dharod.

Writing – review & editing: Mohammed Husein, Carole Debora Nounkeu, Seth Armah, Jigna M. Dharod.

References

1. The Right to Water. UN Human Rights, UN Habitat, World Health Organization. 2010. Available from: <http://www.ohchr.org/Documents/Publications/FactSheet35en.pdf>.
2. Progress on drinking water, sanitation and hygiene in Africa 2000–2020: Five years into the SDGs. New York: United Nations Children's Fund (UNICEF) and World Health Organization (WHO), 2022.
3. Bain R, Cronk R, Hossain R, Bonjour S, Onda K, Wright J, et al. Global assessment of exposure to faecal contamination through drinking water based on a systematic review. *Trop Med Int Health*. 2014 Aug; 19(8):917–27. <https://doi.org/10.1111/tmi.12334> PMID: 24811893
4. Gwimbi P, George M, Ramphalile M. Bacterial contamination of drinking water sources in rural villages of Mohale Basin, Lesotho: exposures through neighbourhood sanitation and hygiene practices. *Environ Health Prev Med*. 2019 May 15; 24(1):33. <https://doi.org/10.1186/s12199-019-0790-z> PMID: 31092211
5. Hamzah L, Boehm AB, Davis J, Pickering AJ, Wolfe M, Mureithi M, et al. Ruminant Fecal Contamination of Drinking Water Introduced Post-Collection in Rural Kenyan Households. *Int J Environ Res Public Health*. 2020 Jan 17; 17(2):608. <https://doi.org/10.3390/ijerph17020608> PMID: 31963600
6. Morgan CE, Bowling JM, Bartram J, Kayser GL. Attributes of drinking water, sanitation, and hygiene associated with microbiological water quality of stored drinking water in rural schools in Mozambique and Uganda. *Int J Hyg Environ Health*. 2021 Jul; 236:113804. <https://doi.org/10.1016/j.ijheh.2021.113804> PMID: 34242999
7. Dadonaite B., Ritchie H., & Roser M. (2018). Diarrheal diseases. Retrieved 6 1, 2022, from Our World in Data. Available from: <https://ourworldindata.org/diarrheal-diseases>.
8. Bain R, Johnston R, Khan S, Hancioglu A, Slaymaker T. Monitoring Drinking Water Quality in Nationally Representative Household Surveys in Low- and Middle-Income Countries: Cross-Sectional Analysis of 27 Multiple Indicator Cluster Surveys 2014–2020. *Environ Health Perspect*. 2021 Sep; 129(9):97010. <https://doi.org/10.1289/EHP8459> PMID: 34546076
9. Ghana Multiple Indicator Cluster Survey (MICS2017/18), Survey Findings Report. Ghana Statistical Service, 2018. Accra, Ghana.
10. UNICEF MICS. MICS6 TOOLS. 2022. Available from: <https://mics.unicef.org/tools#data-collection>.
11. R Core Team R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing. Vienna, Austria. 2020.
12. Progress on household drinking water, sanitation and hygiene 2000–2020: five years into the SDGs. Geneva: World Health Organization (WHO) and the United Nations Children's Fund (UNICEF), 2021. Available from: <https://apps.who.int/iris/handle/10665/345081>.
13. McGuinness SL, O'Toole J, Barker SF, Forbes AB, Boving TB, Giriyan A et al. Household Water Storage Management, Hygiene Practices, and Associated Drinking Water Quality in Rural India. *Environmental Science & Technology*. 2020; 54(8), 4963–4973. <https://doi.org/10.1021/acs.est.9b04818> PMID: 32167297
14. Loyola S, Sanchez JF, Maguiña E, Canal E, Castillo R, Bernal M et al. Fecal Contamination of Drinking Water Was Associated with Diarrheal Pathogen Carriage among Children Younger than 5 Years in Three Peruvian Rural Communities. *Am J Trop Med Hyg*. 2020 Jun; 102(6):1279–1285. <https://doi.org/10.4269/ajtmh.19-0337> PMID: 32207402

15. *Drinking-water*. World Health Organization. 2022. Available from <https://www.who.int/news-room/fact-sheets/detail/drinking-water>.
16. Shields KF, Bain RE, Cronk R, Wright JA, Bartram, J. Association of Supply Type with Fecal Contamination of Source Water and Household Stored Drinking Water in Developing Countries: A Bivariate Meta-analysis. *Environmental Health Perspectives*, 2015: 123(12), 1222–1231.
17. Wright J, Gundry S, Conroy R. Household drinking water in developing countries: a systematic review of microbiological contamination between source and point-of-use. *Trop Med Int Health*. 2004 Jan; 9(1):106–17. <https://doi.org/10.1046/j.1365-3156.2003.01160.x> PMID: 14728614
18. Kandel P, Kunwar R, Lamichhane P, Karki S. Extent of Fecal Contamination of Household Drinking Water in Nepal: Further Analysis of Nepal Multiple Indicator Cluster Survey 2014. *Am J Trop Med Hyg*. 2017 Feb 8; 96(2):446–448. <https://doi.org/10.4269/ajtmh.16-0513> PMID: 27821687
19. Guzmán D, Stoler J. An Evolving Choice in a Diverse Water Market: A Quality Comparison of Sachet Water with Community and Household Water Sources in Ghana. *Am J Trop Med Hyg*. 2018 Aug; 99(2):526–533. <https://doi.org/10.4269/ajtmh.17-0804> PMID: 29893206
20. Chew JF, Corlin L, Ona F, Pinto S, Fenyi-Baah E, Osei BG et al. Water Source Preferences and Water Quality Perceptions among Women in the Eastern Region, Ghana: A Grounded Theory Study. *Int J Environ Res Public Health*. 2019 Oct 11; 16(20):3835. <https://doi.org/10.3390/ijerph16203835> PMID: 31614511
21. Ghana Statistical Service, 2018. Multiple Indicator Cluster Survey (MICS2017/18), Survey Findings Report. Accra, Ghana: GSS.
22. Gundry S, Wright J, Conroy R. A systematic review of the health outcomes related to household water quality in developing countries. *J Water Health*. 2004 Mar; 2(1):1–13. PMID: 15384725
23. Khan JR, Hossain MB, Chakraborty PA, Mistry SK. Household drinking water *E. coli* contamination and its associated risk with childhood diarrhea in Bangladesh. *Environ Sci Pollut Res Int*. 2022 May; 29(21):32180–32189. <https://doi.org/10.1007/s11356-021-18460-9> PMID: 35015232
24. Luby SP, Halder AK, Huda TM, Unicomb L, Islam MS, Arnold BF et al. Microbiological Contamination of Drinking Water Associated with Subsequent Child Diarrhea. *Am J Trop Med Hyg*. 2015 Nov; 93(5):904–911. <https://doi.org/10.4269/ajtmh.15-0274> PMID: 26438031
25. Levy K. Does Poor Water Quality Cause Diarrheal Disease? *Am J Trop Med Hyg*. 2015 Nov; 93(5):899–900. <https://doi.org/10.4269/ajtmh.15-0689> PMID: 26438028
26. Goddard FGB, Pickering AJ, Ercumen A, Brown J, Chang HH, Clasen T. Faecal contamination of the environment and child health: a systematic review and individual participant data meta-analysis. *Lancet Planet Health*. 2020 Sep; 4(9):e405–e415. [https://doi.org/10.1016/S2542-5196\(20\)30195-9](https://doi.org/10.1016/S2542-5196(20)30195-9) PMID: 32918886
27. Demissie GD, Yeshaw Y, Aleminew W, Akalu Y. Diarrhea and associated factors among under five children in sub-Saharan Africa: Evidence from demographic and health surveys of 34 sub-Saharan countries. *PLoS One*. 2021 Sep 20; 16(9):e0257522. <https://doi.org/10.1371/journal.pone.0257522> PMID: 34543347
28. Gakidou E, Cowling K, Lozano R, Murray CJ. Increased educational attainment and its effect on child mortality in 175 countries between 1970 and 2009: a systematic analysis. *Lancet*. 2010 Sep 18; 376(9745):959–74. [https://doi.org/10.1016/S0140-6736\(10\)61257-3](https://doi.org/10.1016/S0140-6736(10)61257-3) PMID: 20851260